



FIRST SEMESTER 2026--2027

COURSE HANDOUT (PART II)

Date: August 3, 2026

In addition to part I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course NO. : BIO G514

Course Title : Molecular Immunology

Instructor In charge : Ashis Kumar Das

1. Course Description: The course will deal extensively with topics like molecular basis of T and B cell antigen recognition and activation. Immunity to microbes and diseases caused by humoral and cell mediated immune responses will be covered and emphasis placed on congenital and acquired immuno-deficiencies. Advanced topics like antibody engineering will be discussed with the help of review articles.

2. Scope and Objectives: The course intends to familiarize the students with modern concepts in immunology. The endeavor will be to highlight concepts in immunology, which would have an influence on understanding the fascinating entity which comprises the body's immune system and ancillary procedure of therapy.

3. Text Book (TB): Cellular and Molecular Immunology, Abbas, Lichtman, Pillai, 10th Ed (2020)., Saunders, Elsevier

4. Reference Books (RB): Kuby Immunology 7th Ed. (2013) Janis Kuby, W.H. Freeman & Co.





Module Number	Lecture Session	Reference (TB)	Learning Objectives
1 Introduction to Immunology Cells and Tissues which comprise the immune system Antibodies and Antigens	L1-3 An overview of Immunology & innate immunity L4-5 Cells and tissues involved in the immune system L6-8 Antibodies and their relationship with antigens	Ch 1,4, 13 Ch 2,3, 13 Ch 5	An exposure to the study of Immunology. Details of cells and tissues involved in vertebrate immunity. Understanding of antibodies and antigens
2 Details of the MHC Presentation & processing of antigens to T lymphocytes Lymphocytes development and maturation Activation of T cells Activation of B cells	L9-11 Major Histocompatibility Complexes L12-14 T Lymphocytes, Antigen recognition & activation L15-17 Somatic recombination of T cell receptor and immunoglobulin genes L18-19T cell activation and signal transduction L20-22 B cell activation and antibody production	Ch 6 Ch 6 Ch 8 Ch7, 9, 10 Ch7, 11	Understanding of presentation of protein antigens to T cells. Phenomena associated with activation along with details of signal transduction pathways. The concepts of diversity arising from the very specific recombination processes and events involved in lymphocyte development and maturation.
3 Immunological tolerance Cytokines Components of complement system	L23-25 T and B cell tolerance L26-27 Role of cytokines in immunity L28-29 Complement system in immunity	Ch 14 Ch 4, 7,10 Ch 4, 12	Understanding of central and peripheral tolerance. Appreciation of cytokines and their actions. Intricacies of the complement system with partly innate and partly adaptive responses.





4 Immunity to microbes Immunity to tumors Hypersensitivity and autoimmunity Antibody engineering and issues related to molecular manipulation of the immune response	L30-32 Microbes and immune responses against them. L33-36 Immune responses against tumors L37-39 Aspects of hypersensitivity and autoimmunity L40-42 Manipulation of components of the immune response mechanism.	Ch 15 Ch 17 Ch 14, 18, 19 Journal articles.	The last module will provide the understanding of immunity to disease causing entities as well as the immune system becoming a disease inflicting mechanism due to breakdowns in the mechanism leading to self injury. Some of the topics will be learnt through group discussions and presentations.
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6. Evaluation Scheme:

Components	Duration	Weightage (%)	Date & Time	Remarks
Mid Sem	90 min	30	10/10	CB
Quizes/ Assignments/ Seminars		30 (10+20)	TBA	CB/OB
Compre Exam	180 min	40	15/12 AN	CB/OB

CB : Closed book, OB: Open book

7. Chamber Consultation Hours: To be announced in the class.

8. Grading Policy:

Award of grades would be guided by the histogram of marks and course average. If a student obtains below 30% of the average of the top three performers or 40% of median, whichever is lower, the student will not be awarded a valid grade and the case will be reported as "NC".

9. Notice: Notices concerning the course will be displayed on Department of Bio. Sciences notice board .

10. Make-up Policy: No make-ups without prior intimation or only in case of hospitalization.





BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
Instruction Division

Instructor-in-charge
BIO G514



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